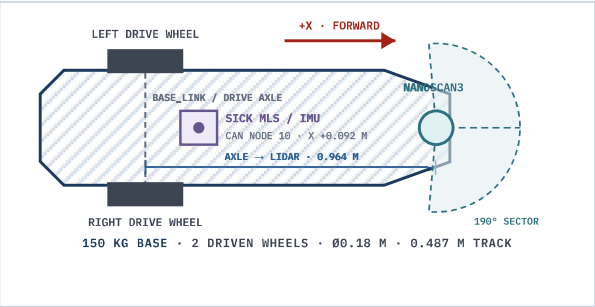




SENSORS & OPERATOR INPUTS

SICK NANOSCAN3	Publishes measurement LaserScan over dedicated Ethernet; its independent OSSD pair goes directly to the FX3.
SICK MLS / IMU	CANopen node 10 supplies timestamped yaw rate; the former magnetic-line tracking function is retired.
MOTOR ENCODERS ×2	BLV-R feedback supplies motor position and velocity; wheel slip and gearbox backlash remain unobserved.
PANEL / PENDANT	Modbus I/O carries Start, Reset, AUTO/MANUAL and four held pendant direction inputs.
CHAFON CF821 RFID	The TCP reader reports tag identity only; it cannot provide a localization pose or precision docking.

ONE INDUSTRIAL PC · N97 · 8 GB

AUTONOMY & APPLICATION

UBUNTU 22.04 · ROS 2 HUMBLE · CYCLONEDDS

AMR SUPERVISOR SLAM / AMCL NAV2 ROUTE CONTROL WEB HMI

Runs one supervised service for live map surveys, immutable revisions, AMCL localization, validated fixed-route missions, process cleanup and operator diagnostics.

ROS 2 TOPICS / TF

BASE & MOTION CONTROL

ROS 2 PROCESSES ON THE SAME INDUSTRIAL PC

AUTHORITY + COMMAND MUX DIFF-DRIVE KINEMATICS WHEEL ODOMETRY

IMU BIAS + EKF

Produces 50 Hz wheel commands under supervisor leases, physical-panel authority and freshness watchdogs. There is no separate motion MCU; each drive closes its own fast current and velocity loops.

HOST POWER

POWER & I/O DISTRIBUTION

Distributes traction and auxiliary power and connects the CANable 2.0 / SLCAN adapter, Ethernet links and Modbus I/O island.

OPERATOR / ENGINEERING CLIENTS

WI-FI / ETHERNET

BROWSER HMI FOXGLOVE SSH / SERVICE

The browser provides mapping, manual jog, route and mission controls. Foxglove and SSH support engineering work. Fleet / MES remains a reserved boundary.

HTTP / DDS

ACTUATION · BLV-R DRIVERS ×2

CANOPEN · CIA 402 PROFILE VELOCITY

NODE 1 · LEFT NODE 2 · RIGHT

Each drive closes its internal current and velocity loops, then returns statusword, velocity and encoder position. Loss of the PC heartbeat forces a drive fault.

CANOPEN

POWER + TORQUE

GEARED WHEEL MOTORS ×2

30:1 GEARBOX · 4,000 R/MIN MOTOR CEILING

Left and right geared motors create differential motion. Their encoders are motor-side, so ground slip and gearbox backlash are not measured directly.

DC / I/O

HWT0 / STO

INDEPENDENT SAFETY PATH nanoScan3 OSSD + safety encoders → SICK Flexi Soft FX3 → HWT01 / HWT02 → drive STO · EDM returns to FX3

SOFTWARE OBSERVES STATUS; IT DOES NOT PERFORM THE SAFETY FUNCTION

